REFLECTIVITY OF THE QUADRATIC FORMS $-d\,x_0^2+x_1^2+\cdots+x_n^2$ OVER $\mathbb{Z}[\sqrt{2}]$: A COMPUTER-ASSISTED CLASSIFICATION

JOHN MCLEOD

Abstract. Let $\mathcal{O}=\mathbb{Z}[\sqrt{2}]$ and let $d\in\mathcal{O}$ be squarefree with $d>0>\sigma(d)$, where σ is the non-trivial Galois automorphism. The form $f_d^n=-d\,x_0^2+x_1^2+\cdots+x_n^2$ then defines a cocompact arithmetic group $O'(f_d^n,\mathcal{O})$ acting on $\mathbb{H}^n$. We study for which (d,n) this group is generated by reflections up to finite index. A monotonicity lemma shows that the reflective n for a fixed d form an interval $\{2,\dots,\mathcal{N}(d)\}$. In particular, f_d^n reflective for some n forces $\langle -d,1,1\rangle$ to be reflective, so it should appear in A. Mark's classification of strongly squarefree reflective rank-3 lattices over $\mathcal{O}$. Mark's table yields 70 candidate values of d. For each of them, and for one further value, we determine $\mathcal{N}(d)$ by our own Vinberg runs together with exactly checkable certificates of non-reflectivity at $\mathcal{N}(d)+1$. There are 27 reflective d, with $\mathcal{N}(d)\in\{2,\dots,6\}$, and 44 values that are non-reflective in every dimension. Six reflective pairs (d,n) appear to be new, including new compact Coxeter polytopes in dimensions 3, 4 and 5. One of them is $d=-1+3\sqrt{2}$, $n=2$: the lattice $\langle 1-3\sqrt{2},1,1\rangle$ is reflective with a compact hexagonal chamber, but it is isometric to no entry of Mark's table, so that table is incomplete. Completeness of our list therefore rests on an explicit, weaker hypothesis about the determinants occurring in Mark's table, which we state and discuss. All decisions use exact arithmetic in $\mathbb{Q}(\sqrt{2})$, and all certificates are re-checkable by an independent verifier.

1. Introduction

Let $K=\mathbb{Q}(\sqrt{2})$ with ring of integers $\mathcal{O}=\mathbb{Z}[\sqrt{2}]$ and embeddings $\mathrm{id}\colon\sqrt{2}\mapsto\sqrt{2}$ and $\sigma\colon\sqrt{2}\mapsto-\sqrt{2}$. We write $\mathrm{N}(x)=x\,\sigma(x)$ and $\varepsilon=1+\sqrt{2}$; then $\mathcal{O}^\times=\pm\varepsilon^{\mathbb{Z}}$ and $\mathrm{N}(\varepsilon)=-1$. Call $d\in\mathcal{O}$ *admissible* if $d>0$ and $\sigma(d)<0$. For admissible d and $n\geq 2$ the form

$$f_d^n=-d\,x_0^2+x_1^2+\cdots+x_n^2$$

has signature $(n,1)$ at id and is positive definite at σ. So $\Gamma_d^n=O'(f_d^n,\mathcal{O})$ is a cocompact arithmetic lattice in $\mathrm{Isom}(\mathbb{H}^n)$ [13, 2]. The lattice $L_d^n=\mathcal{O}^{n+1}$ with this form is *reflective* if the subgroup W_d^n generated by reflections in roots has finite index in Γ_d^n. This is the cocompact analogue of the families $-p\,x_0^2+x_1^2+\cdots+x_n^2$ over $\mathbb{Z}$ studied in [10, 11, 8]. The classical example in our family is $d=\varepsilon$, which is reflective if and only if $n\leq 6$ [4].

Results. We call d *squarefree* if the ideal (d) is.

(1) **Monotonicity** (Lemma 3.1). If f_d^{n+1} is reflective, so is f_d^n. Hence the reflective n for each d form an interval $\{2,\dots,\mathcal{N}(d)\}$, which may be empty.
(2) **Reduction to Mark's table** (§4). If f_d^n is reflective for some n, then $\langle -d,1,1\rangle$ is a reflective strongly squarefree rank-3 lattice. By Mark's theorem [9] it is then isometric to an entry of her table, so (d) is the determinant ideal of an entry. The 522 entries of the table give 70 distinct ideals, hence 70 candidate values of d.
(3) **Unconditional results** (Theorem 7.1). For each of the 70 candidates and one further value, $\mathcal{N}(d)$ is determined exactly (Table 1 and Appendix A): 27 values have $\mathcal{N}(d)\geq 2$ and 44 have $\mathcal{N}(d)<2$. Every claim is backed either by a terminating Vinberg run, cross-checked with CoxIter, or by a certificate of non-reflectivity checked in exact arithmetic.

Date: November 7, 2026.

(4) **A gap in Mark's table** (§8). The lattice $\langle 1-3\sqrt{2},1,1\rangle$ ($d=-1+3\sqrt{2}$, a prime above 17) is reflective with chamber a compact hexagon of angles $(\pi/8,\pi/4,\pi/2)^2$. It is isometric to none of the table entries.
(5) **Completeness, conditionally** (Theorem 7.2). Assume Hypothesis (D): every reflective strongly squarefree rank-3 lattice over $\mathcal{O}$ has the determinant ideal of some table entry. Then Table 1 lists every squarefree admissible d for which f_d^n is reflective for some n. Mark's theorem as stated would imply (D), but it fails at the level of individual lattices. The p17a counterexample does not contradict (D) itself, because its determinant ideal does occur in the table.
(6) **New cases** (§9). Six reflective pairs (d,n) are not in the sources we checked [4, 9, 3, 7]. They are $(-1+3\sqrt{2},2)$, $(\sqrt{2},4)$, $(\sqrt{2},5)$, $(1+2\sqrt{2},4)$, $(3+3\sqrt{2},3)$ and $(-2+3\sqrt{2},3)$. For all d except ε and $-1+2\sqrt{2}$, the upper end $\mathcal{N}(d)$ appears to be new.

Organisation. §2 fixes notation and conventions. §3 proves the monotonicity lemma. §4 gives the reduction from Mark's table to the list of candidates, including how the table was extracted and checked. §5 describes our implementation of Vinberg's algorithm and §6 the certificates of non-reflectivity. §7 states the results, §8 discusses the gap in Mark's table and the hypothesis on which completeness rests, and §9 compares with the literature. §10 lists the calibrations and cross-checks, and §11 explains how to reproduce everything.

2. Setup and conventions

2.1. **Arithmetic in $\mathcal{O}$.** Elements of K are represented exactly as $a+b\sqrt{2}$ with $a,b\in\mathbb{Q}$. The sign of $a+b\sqrt{2}$ is decided exactly: it is the common sign of a and b if they agree, and otherwise it is decided by comparing a^2 with $2b^2$. Every positivity test in this work (inequalities at id, at σ or at both) goes through this function. Floating-point numbers are used only to order batches of candidates and inside one lattice-point enumeration (§5.2), where every output is re-checked exactly; they never accept or reject anything. $\mathcal{O}$ is norm-Euclidean, which gives gcds and factorisation.

2.2. **Canonical parameters.** A unit $u=\pm\varepsilon^k$ is totally positive if and only if $u=\varepsilon^{2j}$, i.e. iff it is a unit square. Multiplying a generator of an ideal by a suitable unit $\pm\varepsilon^k$ gives any sign pattern, because ε has signs $(+,-)$. So the admissible generators of a given ideal form a single orbit $d\,\varepsilon^{2\mathbb{Z}}$. Moreover $x_0\mapsto\varepsilon^k x_0$ shows that f_d^n and $f_{d\varepsilon^{2k}}^n$ are isometric. Hence *admissible squarefree d up to isometry of f_d^n correspond bijectively to squarefree ideals $\mathfrak{a}\neq 0$ of $\mathcal{O}$*. For each ideal we fix the canonical generator $\mathrm{can}(\mathfrak{a})$ that minimises $\max(|d|,|\sigma d|)$; ties are broken by the larger value at id. Throughout, d denotes this canonical generator, and parameters carry short labels: `unit` (ε), `ram2` ($\sqrt{2}$), `inert3` (3ε), `p7a`/`p7b` (the two primes above 7), and `dA_B` or `dmA_B` for $\pm A+B\sqrt{2}$.

For a split rational prime $p=\mathfrak{p}\mathfrak{p}'$ the two primes are genuinely different parameters: if π is admissible so is $-\sigma(\pi)$, which generates $\mathfrak{p}'$, but f_π and $f_{-\sigma\pi}$ are not related by Galois conjugation with respect to the fixed time-like embedding id. The partner $d\mapsto-\sigma(d)$ plays a role in §8.

2.3. **Roots and reflectivity.** A *root* of an $\mathcal{O}$-lattice L is a primitive vector $e\in L$ with totally positive norm $m=(e,e)$ such that $2(e,x)/m\in\mathcal{O}$ for all $x\in L$. The reflection $r_e(x)=x-2\frac{(e,x)}{m}e$ then preserves L. Rescaling e by a unit u gives the same reflection and changes m to u^2m, so norms are taken modulo unit squares. For $L=L_d^n$ with basis $v_0,\dots,v_n$ and $e=\sum k_iv_i$ the conditions read

$$m\mid 2k_i\ (i\geq 1),\qquad m\mid 2d\,k_0,$$

and primitivity forces $m\mid 2d$. Since $\mathrm{N}(\varepsilon)=-1$, every divisor of $2d$ has a totally positive associate. The admissible norms are therefore the totally positive divisors of $2d$ modulo unit squares; for prime $d\neq\sqrt{2}$ these are 1, $2+\sqrt{2}$, 2, d, $(2+\sqrt{2})d$ and $2d$ up to unit squares. Write $W(L)$ for the group generated by reflections in all roots of L and $O'(L)$ for the subgroup of $O(L)$ preserving the

future cone. Then $O'(L) = W(L) \rtimes \mathrm{Sym}(P)$, where P is a chamber of $W(L)$, and L is reflective iff $\mathrm{Sym}(P)$ is finite, iff P has finite volume, iff P has finitely many facets [13].

3. Monotonicity in the dimension

Lemma 3.1 (Monotonicity)**.** *Let L be an admissible hyperbolic lattice over the ring of integers $\mathcal{O}$ of a totally real field. Let $c \in \mathcal{O}$ be totally positive and $L' = L \perp \mathcal{O}u$ with $(u,u) = c$. If L' is reflective, then L is reflective; moreover, a chamber of $W(L)$ is a facet of a chamber of $W(L')$.*

Proof. Write $x \in L'$ as $x = x_L + a(x)u$ with $x_L \in L$ and $a(x) \in \mathcal{O}$, so that

$$(x,u)/c = a(x) \in \mathcal{O} \quad \text{for all } x \in L'. \tag{1}$$

Step 1: the roots of L are exactly the roots of L' orthogonal to u. If r is a root of L of norm m, then $(r,x) = (r,x_L)$ for all $x \in L'$, so $2(r,x)/m \in \mathcal{O}$; and r is primitive in L'. Conversely, a root r' of L' with $r' \perp u$ has $a(r') = 0$ by (1), so it lies in L and is a root there. By (1), u is itself a root of L'.

Step 2. Suppose L' is reflective. Then $W' = W(L')$ has chambers of finite volume and $r_u \in W'$. Choose a chamber P' with a facet $F = P' \cap u^\perp$.

Step 3: no odd angles at F. Suppose a facet with outward root e meets F at angle π/k with k odd. In the dihedral group $\langle r_e, r_u\rangle$ of order $2k$ the reflections r_u and r_e are conjugate, say $w r_u w^{-1} = r_e$. Then w maps the mirror $u^\perp$ to $e^\perp$, so $wu \in \mathbb{R}e$ with $(wu, wu) = c$, and $(wu, u) = \pm c\cos(\pi/k)$. Since $wu \in L'$, (1) gives $\cos(\pi/k) \in \mathcal{O}$. But $2\cos(\pi/k) = \zeta^{-1}(1+\zeta^2)$ with $\zeta = e^{i\pi/k}$. For odd k, $1+\zeta^2 = (1-\zeta^4)/(1-\zeta^2)$ is a cyclotomic unit, so $2\cos(\pi/k)$ is a unit and $\cos(\pi/k)$ is not an algebraic integer. This is a contradiction.

Step 4: every ridge of F lies in a mirror of $W(L)$. Let $R = F \cap F_j$ be a ridge, with angle π/k and k even. If $k = 2$, the root of F_j is orthogonal to u. If $k > 2$ is even, the dihedral group at R contains the reflection in the hyperplane through R perpendicular to $u^\perp$. That reflection lies in W', so its root is a root of L' orthogonal to u. In both cases the root is a root of L by Step 1.

Step 5: F is a chamber of $W(L)$. By Step 1, every mirror of $W(L)$ is of the form $r^\perp \cap u^\perp$ with $r^\perp$ a mirror of W' other than $u^\perp$, so no mirror of $W(L)$ meets the relative interior of F. Hence F lies in one chamber C of $W(L)$. Every ridge of F lies in a mirror of $W(L)$ (Step 4), so $C \subseteq F$ and $C = F$. Since F has finite volume, L is reflective. □

Corollary 3.2. *For admissible d the set $\{n \geq 2 : f_d^n$ reflective$\}$ is an interval $\{2, \ldots, \mathcal{N}(d)\}$, which may be empty. If $f_d^{n_0}$ is not reflective, then neither is f_d^n for any $n \geq n_0$. If f_d^n is reflective for some $n \geq 2$, then $\langle -d, 1, 1\rangle$ is reflective.*

Proof. Apply Lemma 3.1 with $c = 1$ and $u = v_{n+1}$, using $f_d^{n+1} = f_d^n \perp \langle 1\rangle$. □

Remark 3.3. For $c = 1$ this is Bugaenko's face theorem [4] (see also [11, Thm. 3.0.8] and [8, Prop. 3.1] over $\mathbb{Z}$). The proof above also gives a computational check, used in §10: whenever f_d^{n+1} is reflective, the facet $v_{n+1}^\perp$ of its chamber must reproduce the chamber of f_d^n. Its walls come from facets adjacent at angle $\pi/2$, and from projections onto $v_{n+1}^\perp$ of facets adjacent at angle $\pi/4$ or $\pi/8$.

4. Reduction to Mark's table

4.1. **Strongly squarefree lattices.** Following Mark [9, §2], an $\mathcal{O}$-lattice L with discriminant module $\Delta(L) = L^\#/L$ is *strongly squarefree* (SSF) if $\Delta(L)$ needs at most $\frac{1}{2}\operatorname{rank} L$ generators and every invariant factor of $\Delta(L)$ is squarefree.

Lemma 4.1. *If (d) is squarefree, then $\langle -d, 1, 1\rangle$ is strongly squarefree.*

Proof. $L^\#/L \cong \mathcal{O}/d\mathcal{O}$ is cyclic, so it needs $1 \leq 3/2$ generators, and its only invariant factor (d) is squarefree. □

Theorem 4.2 (Mark [9], Thm. 1)**.** *There are* 432 *rank-3 strongly squarefree reflective arithmetic hyperbolic lattices over* $\mathbb{Z}[\sqrt{2}]$*, and they are listed in the tables of* [9, Appendix].

We use the table only through the determinant ideals of its entries. The following elementary lemma connects these ideals to the parameter d.

Lemma 4.3. *Let L be an admissible rank-3 $\mathcal{O}$-lattice with Gram matrix G and $\delta = \det G$. Then $-\delta$ is admissible. If $L \cong \langle -d, 1, 1\rangle$ with d admissible, then $(\delta) = (d)$, and in fact $d = \mathrm{can}((\delta))$ up to unit squares.*

Proof. Signature $(2,1)$ at id gives $\delta < 0$, and positive definiteness at σ gives $\sigma(\delta) > 0$, so $-\delta$ is admissible. An isometry φ with $\varphi^T G' \varphi = G$ has $\det\varphi \in \mathcal{O}^\times$, so $\delta = -d(\det\varphi)^2$. Two admissible generators of the same ideal differ by a totally positive unit, which is a square (§2.2). □

Corollary 4.4 (Reduction)**.** *Let $\mathcal{C} = \{\mathrm{can}((\delta_e)) : e$ an entry of Mark's table$\}$. Assuming Theorem 4.2, if d is squarefree admissible and f_d^n is reflective for some $n \geq 2$, then $d \in \mathcal{C}$.*

Proof. By Corollary 3.2, $\langle -d, 1, 1\rangle$ is reflective. It is SSF by Lemma 4.1, hence isometric to an entry e, and $d = \mathrm{can}((\delta_e))$ by Lemma 4.3. □

Note that the reduction never uses *which* entry $\langle -d, 1, 1\rangle$ is isometric to. Every $d \in \mathcal{C}$ is decided directly by our own computation (§7). So an error in transcribing or matching an individual entry cannot make a wrong d reflective: it could only remove a d from $\mathcal{C}$. The checks below are designed to rule out that failure mode, and §8 discusses the remaining risk, which lies with the table itself.

4.2. **Extracting the table.** The table was read from the TEX source of arXiv:1512.01133v1 (the only arXiv version). The appendix consists of one file per number of sides (3 to 24). Within each file, entries are separated by `\hline\hline`, and each entry gives:

- the determinant δ and its norm;
- a *shape string* listing the angles π/k between consecutive sides of the chamber;
- a symmetric 3×3 Gram matrix, with each element written as a pair $(a\ b)$ for $a + b\sqrt{2}$;
- a list of simple roots with their norms.

A shape string of the form $(s)^2$ indicates a chamber with an order-2 rotational symmetry. Only the roots of one half, $|s|$ of them, are listed.

The parser yields 522 entries. By number of sides they are distributed as follows: 3^3, 4^{35}, 5^{49}, 6^{115}, 7^{33}, 8^{93}, 10^{63}, 11^8, 12^{36}, 14^{14}, 16^{47}, 18^{13}, 20^8, 24^5. Of these, 261 are fully listed and 261 are of the form $(s)^2$. Every entry passes the following exact checks:

(1) G is symmetric with entries in $\mathcal{O}$;
(2) G has signature $(2,1)$ at id and is positive definite at σ;
(3) $\det G = \delta$ exactly, and $\mathrm{N}(\delta)$ equals the printed norm;
(4) (δ) is squarefree, and G is not a scaled lattice (its entries have unit gcd);
(5) each listed vector is primitive, has the stated norm, and satisfies the root condition $2(r,x)/m \in \mathcal{O}$ for the basis vectors x;
(6) the listed roots are pairwise non-obtuse;
(7) the angles between consecutive listed roots, computed from $(r_i, r_j)^2/(m_i m_j) = \cos^2(\pi/k)$, reproduce the shape string.

A first version of the parser misread terms such as `-3w` as $-3 + w$. Check (3) caught this on four entries before any candidate was computed.

4.3. **Candidates.** Applying $e \mapsto \mathrm{can}((\delta_e))$ to the 522 entries gives exactly 70 distinct values of d. They are listed in Appendix A with their factorisations and the number of table entries in each class. The number of entries per class ranges from 1 to 30. The norms $|\mathrm{N}(d)|$ range from 1 to 1666, and composite d (with up to three prime factors) occur. All 16 parameters with $|\mathrm{N}(\mathfrak{p})| \leq 50$ for

prime or unit d (§7) were also computed independently of the table. All of those found reflective at $n = 2$ lie in $\mathcal{C}$. Of the two that are non-reflective at $n = 2$, $5 + 6\sqrt{2}$ is not in $\mathcal{C}$, and $-1 + 4\sqrt{2}$ is in $\mathcal{C}$ because the table contains other, non-diagonal lattices of the same determinant.

4.4. **Duplicates in the table.** The table has 522 entries against the 432 of Theorem 4.2. Grouping the entries by isometry invariants gives 144 groups: determinant ideal, angle cycle up to dihedral symmetry, and multiset of root-norm classes. Within groups we test for isometry by chamber matching (Proposition 8.1); every isometry found is stored as an explicit matrix φ with φ, φ^{-1} integral and $\varphi^T G_B \varphi = G_A$. In the 35 multi-member groups processed so far (179 entries), every group collapsed to a single isometry class. So the table contains at most $522 - 144 = 378$ distinct lattices, and at least 144. No two entries have literally equal Gram matrices. Mark notes that her first-stage lists contain redundancies that are "sorted away in the second step" [9]; the printed table apparently retains some. Duplicates are harmless for Corollary 4.4, which needs each reflective lattice to appear at least once. The remaining 91 groups have not been processed.

5. Vinberg's algorithm over $\mathcal{O}$

5.1. **The diagonal forms f_d^n.** We use Vinberg's algorithm [13] in Bugaenko's form for totally real fields, with control vector v_0.

(1) *Stabiliser.* All roots with $k_0 = 0$ are roots of the positive definite lattice $\mathcal{O}^n$. We enumerate them for every admissible norm and check that the simple system at v_0 is of type B_n: $v_{i+1} - v_i$ for $1 \le i < n$, and $-v_n$. Over $\mathcal{O}$ this is not automatic, and the code verifies it rather than assuming it.

(2) *Batches.* Candidate roots e with $k_0 > 0$ at id are processed in batches of equal distance $\delta(e) = d\,k_0^2/m$, which is $\sinh^2$ of the distance from v_0 to the mirror. Ties are decided exactly in K. For fixed (m, k_0) the conjugate equation $\sum \sigma(k_i)^2 = \sigma(m) - \sigma(d)\sigma(k_0)^2 \ge 0$ bounds $|\sigma(k_0)|$, so each batch is finite.

(3) *Solving $\sum_{i \ge 1} k_i^2 = m + d\,k_0^2$.* We run a depth-first search over $k_1 \ge \cdots \ge k_n \ge 0$ at id, carrying the residuals at both embeddings, which bound each k_i in a box. Partial assignments violating $m \mid 2k_i$ are pruned. We then require the gcd of the coordinates to be a unit, and normalise each root to the fixed representative of its norm class.

(4) *Acceptance.* A candidate is accepted iff it is non-obtuse at id to every previously accepted root. That includes roots accepted earlier in the same batch; survivors of one batch are asserted, not assumed, to be mutually non-obtuse.

(5) *Termination.* The polytope cut out by the accepted roots is compact iff it has a vertex and every elliptic subdiagram of rank $n - 1$ extends to exactly two elliptic subdiagrams of rank n [13]. Ellipticity is tested by exact leading principal minors of the raw Gram matrix $((e_i, e_j))$, which is congruent to the normalised one. Edge labels are obtained by comparing $(e_i, e_j)^2/(m_i m_j)$ exactly with $\cos^2(\pi/k)$ for $k = 2, 3, 4, 6, 8$, or recognising it as > 1 (ultraparallel). Any other value raises an error. So does the value 1 (parallel), which is impossible for a cocompact group.

(6) *Caps.* Each run has limits on δ, on the number of roots and on wall-clock time. A run that reaches a cap is recorded as *undetermined* and passed to the certificate search; it is never counted as non-reflective.

Every terminating run is cross-checked with CoxIter [5], which confirms compactness and finite covolume and gives the f-vector and Euler characteristic.

5.2. **General rank-3 lattices.** To compute chambers of arbitrary entries of Mark's table we wrote a second, independent implementation for any rank-3 Gram matrix G over $\mathcal{O}$. It shares no enumeration code with the diagonal one.

- *Root norms.* Root norms m divide $2 \det G$, because $2r/m \in L^\#$ and $(\det G)L^\# \subseteq L$.

- *Enumeration as a lattice-point problem.* For a timelike control vector v_0, roots at distance $\leq \Delta$ satisfy two conditions. At id, $H(r) = Q(r) - 2(r, v_0)^2/Q(v_0) \leq m(1 + 2\Delta)$, where H is the positive majorant attached to v_0. At σ, $\sigma Q(r) = \sigma(m)$. So they are points of $\mathbb{Z}^6$ in an explicit ellipsoid.
- *Fincke–Pohst.* We enumerate that ellipsoid after an exact rational LLL reduction, using a 40-digit rational approximation of $\sqrt{2}$ only to build the ellipsoid, with a safety margin. Every candidate is re-checked exactly.

The second implementation reproduces the chambers of the diagonal code (§8) and of the fully listed table entries we tested. It also completes the half-listed $(s)^2$ chambers.

6. Certificates of non-reflectivity

If a run does not terminate, we look for $g \in O'(L)$ of infinite order with $g(P) = P$. Then $\mathrm{Sym}(P)$ is infinite and L is not reflective.

6.1. **Construction.** Let S_0 be the stabiliser simple system at v_0, whose roots are facets of P. Suppose $S' \subseteq R$ is a set of found facet roots that is elliptic of rank n, with the same Gram matrix as S_0 under a bijection. Together with the vertex vectors, this determines a linear map h with $h(S_0) = S'$. If h is integral and preserves the form, then h maps the cone of P at its vertex v_0 onto the cone of P at the vertex v' cut out by S'. So $h(P)$ and P share interior points. Since h normalises W, $h(P) = P$. Such vertex symmetries h may have finite order. A certificate is a product $g = h_1 \cdots h_k$ (in practice $k \leq 2$) that has infinite order. For one case ($5 + 6\sqrt{2}$, $n = 2$) the base vertex is another found elliptic set rather than S_0; the verifier then re-establishes both ends as vertices of P.

6.2. **Exact checks.** For each certificate, `verify_certificate.py` checks:

(1) every entry of every h_i lies in $\mathcal{O}$, and $h_i^T F h_i = F$;
(2) $(h_i v_0, v_0) < 0$ at id, so $h_i \in O'$;
(3) h_i maps the listed simple system to the listed target system, which is elliptic of rank n, and every listed vector is a root of L;
(4) every target root is a facet of P. This cannot be read off from the certificate alone, so the verifier re-runs the deterministic Vinberg prefix up to the largest distance involved and confirms that the roots occur in it;
(5) g has infinite order. The verifier exhibits a real eigenvalue $\lambda > 1$ of g at id: a rational interval $[a, b]$ with $1 < a$ on which the characteristic polynomial χ_g changes sign, evaluated exactly. Equivalently, $\mathrm{N}(\chi_g) \in \mathbb{Z}[x]$ has a non-cyclotomic factor.

The step "an elliptic rank-n set of facets of P meets at a vertex whose cone is cut out by exactly those facets" is Vinberg's for acute-angled polyhedra. For a chamber with infinitely many facets, local finiteness of the mirror arrangement reduces it to the finite case. We record this as the one step of the certificate argument that rests on a standard fact rather than on an explicit computation.

6.3. **Example.** For $d = -1 + 4\sqrt{2}$ (a prime above 31) and $n = 2$, the run does not terminate. The certificate is the product of two vertex symmetries,

$$g = \begin{pmatrix} 712 + 504\sqrt{2} & -328 - 232\sqrt{2} & -37 - 26\sqrt{2} \\ 1093 + 774\sqrt{2} & -504 - 356\sqrt{2} & -56 - 40\sqrt{2} \\ 1080 + 764\sqrt{2} & -497 - 352\sqrt{2} & -56 - 40\sqrt{2} \end{pmatrix},$$

with $\chi_g(x) = x^3 - (152 + 108\sqrt{2})x^2 + (152 + 108\sqrt{2})x - 1$. This polynomial has a root in $[607/2, 304]$, so g is loxodromic.

In total 167 certificates were produced and verified: 111 for the 16×9 grid of §7.1 and the rest for the classification. Table 4 in Appendix A summarises the one used at $\mathcal{N}(d) + 1$ for each d.

7. Results

7.1. **The grid $|\mathrm{N}(\mathfrak{p})| \le 50$, $2 \le n \le 10$.** Independently of Mark's table, we first decided every pair (d, n) with d the unit ε or a prime with $|\mathrm{N}(\mathfrak{p})| \le 50$, and $2 \le n \le 10$. That covers the ramified prime $(\sqrt{2})$, the inert primes $3, 5$, and the two primes above each of $7, 17, 23, 31, 41, 47$: 16 parameters and 144 pairs. Each pair is decided by a terminating run, by a certificate at that n, or (for 5 pairs) by Corollary 3.2 from a certificate at a lower n. These results are included in Table 1.

7.2. **The classification.**

Theorem 7.1 (unconditional)**.** *For each d in Table 1, f_d^n is reflective exactly for $2 \le n \le \mathcal{N}(d)$, with the stated number of facets, and is non-reflective for every $n > \mathcal{N}(d)$. For each of the remaining 44 values of d in Appendix A (the other candidates, and $-1 + 12\sqrt{2}$), f_d^n is non-reflective for every $n \ge 2$.*

Proof. Reflectivity for $2 \le n \le \mathcal{N}(d)$ follows from terminating runs, each confirmed compact by CoxIter. Non-reflectivity at $\mathcal{N}(d) + 1$ follows from the certificates of §6, and at all larger n by Corollary 3.2. □

Theorem 7.2 (conditional)**.** *Assume Hypothesis (D) of §8. Then Table 1 is the complete list of squarefree admissible d for which f_d^n is reflective for some $n \ge 2$. For every other squarefree admissible d, f_d^n is non-reflective for all $n \ge 2$.*

Proof. By Corollary 3.2 and Lemmas 4.1–4.3, reflectivity of some f_d^n makes $\langle -d, 1, 1\rangle$ a reflective SSF lattice with determinant ideal (d). Under (D), (d) is the determinant ideal of a table entry, i.e. $d \in \mathcal{C}$. All of $\mathcal{C}$ is decided by Theorem 7.1. □

Table 1: The 27 reflective d. Columns: label, d, $\mathrm{N}(d)$, factorisation of (d), reflective range, number of facets for $n = 2, 3, \ldots$, first non-reflective $n_0 = \mathcal{N}(d) + 1$ (each with a certificate), and the source of each reflective n in order ([9] at $n = 2$ means an explicit isometry to a table entry). **new**: not in the sources checked (§9).

label	d	$\mathrm{N}(d)$	(d)	n	facets	n_0	sources
`unit`	$1+\sqrt{2}$	-1	unit	2–6	3, 5, 7, 10, 34	7 [4]	[4], [4], [4], [4], [4]
`ram2`	$\sqrt{2}$	-2	$(\sqrt{2})$	2–5	4, 6, 8, 27	6	[9], [3], **new**, **new**
`p7a`	$-1+2\sqrt{2}$	-7	$(3+\sqrt{2})$	2–5	4, 5, 8, 20	6 [7]	[9], [3], [7], [7]
`p7b`	$1+2\sqrt{2}$	-7	$(3-\sqrt{2})$	2–4	4, 6, 9	5	[9], [3], **new**
`inert3`	$3+3\sqrt{2}$	-9	(3)	2–3	4, 12	4	[9], **new**
`dm2_3`	$-2+3\sqrt{2}$	-14	$(\sqrt{2})\cdot(3-\sqrt{2})$	2–3	5, 18	4	[9], **new**
`p23a`	$-3+4\sqrt{2}$	-23	$(5+\sqrt{2})$	2–3	5, 10	4	[9], [3]
`d2_3`	$2+3\sqrt{2}$	-14	$(\sqrt{2})\cdot(3+\sqrt{2})$	2–2	5	3	[9]
`p17a`	$-1+3\sqrt{2}$	-17	$(5+2\sqrt{2})$	2–2	6	3	**new**
`p17b`	$1+3\sqrt{2}$	-17	$(5-2\sqrt{2})$	2–2	6	3	[9]
`d0_3`	$3\sqrt{2}$	-18	$(\sqrt{2})\cdot(3)$	2–2	6	3	[9]
`p23b`	$3+4\sqrt{2}$	-23	$(5-\sqrt{2})$	2–2	6	3	[9]
`inert5`	$5+5\sqrt{2}$	-25	(5)	2–2	6	3	[9]
`p31b`	$1+4\sqrt{2}$	-31	$(7-3\sqrt{2})$	2–2	8	3	[9]
`p41a`	$-3+5\sqrt{2}$	-41	$(7+2\sqrt{2})$	2–2	12	3	[9]
`p41b`	$3+5\sqrt{2}$	-41	$(7-2\sqrt{2})$	2–2	12	3	[9]
`p47a`	$-5+6\sqrt{2}$	-47	$(7+\sqrt{2})$	2–2	10	3	[9]
`d7_7`	$7+7\sqrt{2}$	-49	$(3+\sqrt{2})\cdot(3-\sqrt{2})$	2–2	5	3	[9]
`dm6_7`	$-6+7\sqrt{2}$	-62	$(\sqrt{2})\cdot(7-3\sqrt{2})$	2–2	11	3	[9]
`d3_6`	$3+6\sqrt{2}$	-63	$(3)\cdot(3-\sqrt{2})$	2–2	8	3	[9]
`d0_7`	$7\sqrt{2}$	-98	$(\sqrt{2})\cdot(3+\sqrt{2})\cdot(3-\sqrt{2})$	2–2	8	3	[9]

label	d	N(d)	(d)	n	facets	n_0	sources
d9_10	$9+10\sqrt{2}$	-119	$(3+\sqrt{2})\cdot(5-2\sqrt{2})$	2–2	8	3	[9]
dm9_10	$-9+10\sqrt{2}$	-119	$(3-\sqrt{2})\cdot(5+2\sqrt{2})$	2–2	12	3	[9]
dm6_9	$-6+9\sqrt{2}$	-126	$(\sqrt{2})\cdot(3)\cdot(3-\sqrt{2})$	2–2	11	3	[9]
d1_9	$1+9\sqrt{2}$	-161	$(3-\sqrt{2})\cdot(5-\sqrt{2})$	2–2	18	3	[9]
dm1_9	$-1+9\sqrt{2}$	-161	$(3+\sqrt{2})\cdot(5+\sqrt{2})$	2–2	18	3	[9]
d1_12	$1+12\sqrt{2}$	-287	$(3+\sqrt{2})\cdot(7+2\sqrt{2})$	2–2	24	3	[9]

7.3. **Polytopes.** Table 2 gives the CoxIter invariants of all reflective cases, and the angle cycle for $n = 2$. Some combinatorial types stand out:

- $(\sqrt{2}, 4)$ has the f-vector of the 4-cube;
- $(3+3\sqrt{2}, 3)$ has the f-vector of the dodecahedron;
- the largest polygon is the 24-gon for $d = 1+12\sqrt{2}$;
- the largest polytope is the 34-facet polytope for $(\varepsilon, 6)$.

Figure 1 shows the Coxeter diagrams of the new cases.

Table 2: Invariants of the reflective chambers (CoxIter; all are compact): facets, f-vector, Euler characteristic, and for $n = 2$ the cyclic sequence of angles π/k (minimal under rotation and reflection). ⋆: new case.

label	n	facets	f-vector	χ	angles	
unit	2	3	(3, 3, 1)	$-1/16$	248	
unit	3	5	(6, 9, 5, 1)	0		
unit	4	7	(13, 26, 20, 7, 1)	11/768		
unit	5	10	(30, 75, 76, 39, 10, 1)	0		
unit	6	34	(530, 1590, 1850, 1050, 294, 34, 1)	$-3971/15360$		
ram2	2	4	(4, 4, 1)	$-1/8$	2224	
ram2	3	6	(8, 12, 6, 1)	0		
ram2	4	8	(16, 32, 24, 8, 1)	11/384		⋆
ram2	5	27	(150, 375, 350, 150, 27, 1)	0		⋆
p7a	2	4	(4, 4, 1)	$-1/8$	2224	
p7a	3	5	(6, 9, 5, 1)	0		
p7a	4	8	(15, 30, 23, 8, 1)	55/1152		
p7a	5	20	(96, 240, 228, 102, 20, 1)	0		
p7b	2	4	(4, 4, 1)	$-1/4$	2424	
p7b	3	6	(8, 12, 6, 1)	0		
p7b	4	9	(18, 36, 27, 9, 1)	55/576		⋆
inert3	2	4	(4, 4, 1)	$-5/16$	2248	
inert3	3	12	(20, 30, 12, 1)	0		⋆
dm2_3	2	5	(5, 5, 1)	$-3/8$	22224	
dm2_3	3	18	(32, 48, 18, 1)	0		⋆
p23a	2	5	(5, 5, 1)	$-11/24$	22234	
p23a	3	10	(16, 24, 10, 1)	0		
d2_3	2	5	(5, 5, 1)	$-3/8$	22224	
p17a	2	6	(6, 6, 1)	$-9/8$	248248	⋆
p17b	2	6	(6, 6, 1)	$-9/8$	248248	
d0_3	2	6	(6, 6, 1)	$-5/8$	222224	
p23b	2	6	(6, 6, 1)	$-11/12$	234234	
inert5	2	6	(6, 6, 1)	$-13/16$	224228	
p31b	2	8	(8, 8, 1)	$-5/4$	22242224	
p41a	2	12	(12, 12, 1)	$-21/8$	224228224228	

label	n	facets	f-vector	χ	angles
p41b	2	12	(12, 12, 1)	$-21/8$	224228224228
p47a	2	10	(10, 10, 1)	$-23/12$	2223422234
d7_7	2	5	(5, 5, 1)	$-9/16$	22248
dm6_7	2	11	(11, 11, 1)	$-15/8$	22222222224
d3_6	2	8	(8, 8, 1)	$-5/4$	22242224
d0_7	2	8	(8, 8, 1)	$-9/8$	22222224
d9_10	2	8	(8, 8, 1)	$-9/8$	22222224
dm9_10	2	12	(12, 12, 1)	$-9/4$	222224222224
dm6_9	2	11	(11, 11, 1)	$-15/8$	22222222224
d1_9	2	18	(18, 18, 1)	$-33/8$	222222248222222248
dm1_9	2	18	(18, 18, 1)	$-33/8$	222224228222224228
d1_12	2	24	(24, 24, 1)	$-21/4$	22222222224224222222224

7.4. **Observations.**

- $\mathcal{N}(d)$ falls quickly with $|\mathrm{N}(d)|$. Only the seven values with $|\mathrm{N}(d)| \le 23$ listed first in Table 1 reach $n \ge 3$, and none reaches $n \ge 7$. This is far below the non-cocompact pattern over $\mathbb{Q}$, where $d = 1, 2, 3$ reach $n = 19, 14, 13$.
- The two primes above a split p can behave differently: $p = 7$ ($\mathcal{N} = 5$ against 4), $p = 23$ (3 against 2), $p = 31$ ($\emptyset$ against 2). Conjugate partners $d \leftrightarrow -\sigma(d)$ are not equivalent (§2.2).
- Among the 70 candidates, 44 minus one are non-reflective already at $n = 2$: for them the table only contains non-diagonal lattices of determinant (d).

8. The gap in Mark's table and Hypothesis (D)

8.1. **Isometry testing by chamber matching.**

Proposition 8.1. *Let L_A, L_B be reflective rank-3 lattices with chambers P_A, P_B and simple roots (a_i) and (b_j). If $\varphi\colon L_A \to L_B$ is an isometry, then there is $w \in W(L_B)$ such that $w\varphi$ maps each a_i to $u_i b_{\pi(i)}$ for an injection π and units $u_i > 0$ at* id *with $u_i^2 = (a_i, a_i)/(b_{\pi(i)}, b_{\pi(i)})$.*

Proof. φ conjugates $W(L_A)$ onto $W(L_B)$, so $\varphi(P_A)$ is a chamber of $W(L_B)$ and some $w \in W(L_B)$ maps it to P_B. Then $w\varphi$ maps facets to facets. A facet determines its primitive outward root up to a positive unit. □

Hence, given the *complete* simple-root cycle of L_B, enumerating the injections compatible with the Gram matrices decides isometry. Any φ obtained by solving on three independent roots is accepted only if φ and φ^{-1} are integral and $\varphi^T G_B \varphi = G_A$. For each reflective $d \in \mathcal{C}$ we tested $\langle -d, 1, 1\rangle$ against every entry with determinant ideal (d). Every one of them except $d = -1 + 3\sqrt{2}$ is isometric to at least one entry (last column of Table 3).

8.2. **The lattice $\langle 1 - 3\sqrt{2}, 1, 1\rangle$.** Let $d = -1 + 3\sqrt{2}$, so $\mathrm{N}(d) = -17$ and (d) is one of the primes above 17. Vinberg's algorithm terminates with six roots. In cyclic order they are

$$\begin{aligned}
e_1 &= (13 + 9\sqrt{2},\ 17 + 12\sqrt{2},\ 16 + 11\sqrt{2}), & (e_1, e_1) &= 2 + \sqrt{2},\\
e_2 &= (0, -1, 1), & (e_2, e_2) &= 2,\\
e_3 &= (0, 0, -1), & (e_3, e_3) &= 1,\\
e_4 &= (3 + 2\sqrt{2},\ 5 + 4\sqrt{2},\ 0), & (e_4, e_4) &= 2 + \sqrt{2},\\
e_5 &= (2 + \sqrt{2},\ 3 + 2\sqrt{2},\ 1 + \sqrt{2}), & (e_5, e_5) &= 2,\\
e_6 &= (4 + 3\sqrt{2},\ 6 + 4\sqrt{2},\ 5 + 3\sqrt{2}), & (e_6, e_6) &= 1,
\end{aligned}$$

with consecutive inner products $(e_1, e_2) = (e_4, e_5) = -1 - \sqrt{2}$, $(e_2, e_3) = (e_5, e_6) = -1$ and $(e_3, e_4) = (e_6, e_1) = 0$. The chamber is therefore a compact hexagon with angles $(\pi/8, \pi/4, \pi/2, \pi/8, \pi/4, \pi/2)$;

CoxIter gives $\chi = -9/8$. The independent rank-3 implementation of §5.2 was started from an interior point of this hexagon. It also terminates with exactly these six roots.

Mark's table has 8 entries with determinant ideal $(-1 + 3\sqrt{2})$. Their shapes are $(322)^2$ (two entries) and $(2222)^2$ (six entries); none has angle cycle 842842. The angle cycle of the chamber is an isometry invariant, so none of these entries is isometric to $\langle -d, 1, 1\rangle$. This argument relies on the printed shapes. To remove that dependence, we recomputed the full chambers of all 8 entries with the implementation of §5.2. Each run terminates and contains the printed roots. The two $(322)^2$ entries are hexagons with angles $(\pi/3, \pi/2, \pi/2)^2$, and the six $(2222)^2$ entries are octagons. So none of them is an 842842 hexagon, independently of the printed data. The only hexagon of shape $(842)^2$ in the whole table has determinant ideal $(1 + 3\sqrt{2})$, the other prime above 17. That entry is explicitly isometric to $\langle -(1 + 3\sqrt{2}), 1, 1\rangle$, and the two hexagons have identical angles and root norms.

Proposition 8.2. *$\langle 1 - 3\sqrt{2}, 1, 1\rangle$ is a reflective strongly squarefree rank-3 lattice over $\mathcal{O}$ that is isometric to no entry of the table in* [9].

We do not know the cause. One possibility is that the enumeration identified the two hexagons over the conjugate primes above 17, which have identical combinatorics and norms. Another is an omission at some stage of the enumeration or of typesetting. The published version of [9] has not been compared with the arXiv source.

8.3. **What completeness rests on.** Proposition 8.2 shows that Theorem 4.2 cannot be used as stated. Corollary 4.4 only needs the weaker statement:

Hypothesis (D). *For every reflective strongly squarefree rank-3 lattice L over $\mathbb{Z}[\sqrt{2}]$, the ideal $(\det L)$ is the determinant ideal of some entry of Mark's table, i.e. belongs to the 70 ideals of Appendix A.*

Proposition 8.2 does not contradict (D), because $(-1 + 3\sqrt{2})$ does occur in the table. But it shows that the table can miss a reflective lattice whose determinant ideal is otherwise represented. Nothing excludes an omission whose ideal is not represented. Two partial safeguards:

(1) *Conjugate partners.* If the gap arose from identifying conjugate primes, its most likely trace is a reflective d whose partner $-\sigma(d)$ is not a candidate. We computed $-\sigma(d)$ for each of the 27 reflective d. All are candidates except $5 + 6\sqrt{2}$ (partner of $-5 + 6\sqrt{2}$), which is certified non-reflective at $n = 2$ in the grid, and $-1 + 12\sqrt{2}$ (partner of $1 + 12\sqrt{2}$). We decided $-1 + 12\sqrt{2}$ directly: it is non-reflective at $n = 2$, with a certificate. Partners are not equivalent in general: for example, $1 + 4\sqrt{2}$ is reflective and $-1 + 4\sqrt{2}$ is not. So this check shows only that no partner-type omission occurs among our reflective d.
(2) *Independent grid.* For all 16 prime or unit parameters with $|\mathrm{N}(\mathfrak{p})| \le 50$ our own computation agrees with membership in $\mathcal{C}$ (§4.2).

Proving (D) unconditionally would need an independent rank-3 classification, or a re-verification of the relevant part of [9], for example of the Allcock–Nikulin type bounds on short and close pairs of roots. We have not attempted this.

9. Comparison with the literature

Every reflective pair (d, n) was compared with the following sources, after checking isometry explicitly.

- $n = 2$. An explicit isometry to a table entry of [9]; this covers every case except $d = -1 + 3\sqrt{2}$.
- $d = \varepsilon$. Reflective iff $n \le 6$ [4], as cited in [7, 2]; we did not have access to [4] itself.
- $n = 3$. Bogachev's list of candidate lattices $L(1)$–$L(11)$ in [3]:

  - the diagonal rows $L(1), L(2), L(3), L(5)$ are $d = \varepsilon$, $1 + 2\sqrt{2}$, $-1 + 2\sqrt{2}$ (as $5 + 4\sqrt{2} = (-1 + 2\sqrt{2})\varepsilon^2$) and $\sqrt{2}$;
  - $L(8) = \langle -(7+6\sqrt{2}), 1, 1, 1\rangle$ is $d = -3+4\sqrt{2}$, since $7+6\sqrt{2} = (-3+4\sqrt{2})\varepsilon^2$. Bogachev shows it is reflective but not sub-2-reflective;
  - his non-diagonal rows $L(6), L(7), L(10), L(11)$ have determinant ideals $(2+3\sqrt{2})$, $(2+3\sqrt{2})$, $(3+4\sqrt{2})$ and $(11+12\sqrt{2})$. So they cover neither $3+3\sqrt{2}$ nor $-2+3\sqrt{2}$.
- $d = -1 + 2\sqrt{2}$. Reflective for $n \le 5$ by Guglielmetti [7], via $\langle 1 - 2\Theta_2, 1, 1, 1, 1\rangle$ and $\langle -5 - 4\Theta_2, 1, \ldots, 1\rangle$ with $\Theta_2 = \sqrt{2}$. Our $n = 5$ polytope has 20 facets and f-vector $(96, 240, 228, 102, 20, 1)$, as in [7].

The remaining reflective pairs are, as far as these sources go, new:

$$(-1 + 3\sqrt{2}, 2), \quad (\sqrt{2}, 4), \quad (\sqrt{2}, 5), \quad (1 + 2\sqrt{2}, 4), \quad (3 + 3\sqrt{2}, 3), \quad (-2 + 3\sqrt{2}, 3).$$

The upper ends $\mathcal{N}(d)$ were known before only for ε [4] and, in effect, for $-1 + 2\sqrt{2}$ [7]. For the candidates that are non-reflective at $n = 2$, non-reflectivity at $n = 2$ also follows from [9].

Remark 9.1 (Non-reduced coefficients)**.** Bogachev's rows $L(4) = \langle -(11+8\sqrt{2}), 1, 1, 1\rangle$ and $L(9) = \langle -(7+5\sqrt{2}), 1, 1, 1\rangle$ are isometric to $L(2)$ and $L(1)$, since $11+8\sqrt{2} = (1+2\sqrt{2})\varepsilon^2$ and $7+5\sqrt{2} = \varepsilon^3$. Their printed root lists nevertheless differ from those of $L(2)$ and $L(1)$. AlVin reproduces the printed lists. In both cases the root $(\sqrt{2}-1, 1+\sqrt{2}, 0, 0)$ should be accepted first and is missed. The likely cause is that candidates with $0 < k_0 < 1$ at id are skipped, which matters only when the coefficient is not reduced modulo unit squares. Our implementation gives the same chamber for d and $d\varepsilon^{2k}$; this is tested. Bogachev's conclusions about $L(4)$ and $L(9)$ are unaffected, because those lattices are isometric to $L(2)$ and $L(1)$.

10. Calibration and cross-checks

The same code, with the ring interface instantiated to $\mathbb{Z}$, reproduces the following.

- *Over $\mathbb{Q}$:* $p = 3$ reflective iff $n \le 13$ [10]; $p = 5$ iff $n \le 8$ [8]; and the maxima $3, 4, 2, 3, 2, 2$ for $d = 7, 11, 13, 17, 19, 23$ of [11, Table 3.1].
- *$d = \varepsilon$:* terminates for $n \le 6$ and not at $n = 7$ [4]. Our certificate at $n = 7$ makes the latter rigorous.
- *$d = -1 + 2\sqrt{2}$:* the facet count and f-vector of Guglielmetti's $n = 5$ polytope [7].
- *AlVin:* identical root sets for nine pairs (d, n) over $\mathbb{Z}$ and $\mathcal{O}$.
- *CoxIter:* compactness, f-vector and Euler characteristic of every reflective case.

In addition:

- *Facet test* (from Lemma 3.1): for all 14 grid pairs where both (d, n) and $(d, n+1)$ are reflective, the facet $v_{n+1}^{\perp}$ of the $(d, n+1)$ chamber reproduces the (d, n) chamber exactly.
- *Rescaling:* d and $d\varepsilon^{2k}$ give the same chamber (Remark 9.1).
- *The table:* the seven exact checks of §4.2 on all 522 entries.

Finally, for each certified case of the grid we searched for a reflective super-lattice $L' \supset L$ in which the infinite-order symmetry becomes a product of reflections (cf. [11, Fig. 3.2]). None was found within the caps used. This is recorded as "not found", not as a negative result.

11. Reproducibility

The software is written in Python, with exact arithmetic in `fractions.Fraction`. It is available at `[URL to be added]`. The ancillary files of this submission contain:

- the verifier `verify_certificate.py` together with the package `reflect/`;
- all 167 certificates;
- for every reflective case, the roots, Gram matrix and Coxeter diagram;
- the per-d summary `classification.json`;

- the stored isometries to table entries;
- the parser for Mark's TEX source. The table itself is not redistributed and is regenerated from arXiv:1512.01133.

The test suite (913 tests) re-verifies every certificate and stored isometry, the table checks, the calibrations and the literature attributions. One uncertain step is isolated in §6: elliptic subdiagrams correspond to vertices for chambers with infinitely many facets.

Acknowledgements. The software, the computations and a first draft of this report were produced with the assistance of Claude (Anthropic), an AI system, under the author's direction. The author has reviewed the mathematics and is responsible for the content. We thank the authors of AlVin and CoxIter for making their software available.

References

[1] D. Allcock, *The reflective Lorentzian lattices of rank 3*, Mem. Amer. Math. Soc. **220** (2012), no. 1033.
[2] M. Belolipetsky, *Arithmetic hyperbolic reflection groups*, Bull. Amer. Math. Soc. **53** (2016), 437–475; arXiv:1506.03111.
[3] N. Bogachev, *From geometry to arithmetic of compact hyperbolic Coxeter polytopes*, arXiv:2003.11944.
[4] V. O. Bugaenko, Selecta Math. Soviet. **9** (1990).
[5] R. Guglielmetti, *CoxIter – computing invariants of hyperbolic Coxeter groups*, LMS J. Comput. Math. **18** (2015), 754–773.
[6] R. Guglielmetti, *AlVin: a C++ implementation of the Vinberg algorithm for diagonal quadratic forms*, `https://github.com/rgugliel/AlVin`.
[7] R. Guglielmetti, *Hyperbolic isometries in (in-)finite dimensions and discrete reflection groups: theory and computations*, PhD thesis, Université de Fribourg, 2017.
[8] A. Mark, *Reflection groups of the quadratic form $-px_0^2 + x_1^2 + \cdots + x_n^2$ with p prime*, Publ. Mat. **59** (2015), 353–372; arXiv:1207.4822.
[9] A. Mark, *The classification of rank 3 reflective hyperbolic lattices over $\mathbb{Z}[\sqrt{2}]$*, Math. Proc. Cambridge Philos. Soc. **164** (2018), 221–257; arXiv:1512.01133.
[10] J. Mcleod, *Hyperbolic reflection groups associated to the quadratic forms $-3x_0^2 + x_1^2 + \cdots + x_n^2$*, arXiv:1007.2299.
[11] J. Mcleod, *Arithmetic hyperbolic reflection groups*, PhD thesis, Durham University, 2013.
[12] V. V. Nikulin, *On the classification of hyperbolic root systems of rank three*, arXiv:alg-geom/9711032.
[13] E. B. Vinberg, *On groups of unit elements of certain quadratic forms*, Math. USSR Sb. **16** (1972), 17–35.
[14] E. B. Vinberg, *The absence of crystallographic groups of reflections in Lobachevsky spaces of large dimension*, Trans. Moscow Math. Soc. (1985), 75–112.

Appendix A. All candidates

Table 3: The 70 candidates $d = \mathrm{can}((\delta_e))$ from Mark's table, and $-1 + 12\sqrt{2}$ (†, conjugate partner of $1 + 12\sqrt{2}$, not a candidate). Columns: label, d, $\mathrm{N}(d)$, factorisation, number of table entries with determinant ideal (d), result at $n = 2$ (R: reflective, with number of facets; N: non-reflective, with the number of roots found before the certificate search), and for reflective d the number of those entries isometric to $\langle -d, 1, 1\rangle$.

label	d	$\mathrm{N}(d)$	(d)	entries	$n = 2$	isometric
`unit`	$1 + \sqrt{2}$	-1	unit	1	R (3)	1/1
`ram2`	$\sqrt{2}$	-2	$(\sqrt{2})$	2	R (4)	1/2
`p7a`	$-1 + 2\sqrt{2}$	-7	$(3 + \sqrt{2})$	10	R (4)	6/10
`p7b`	$1 + 2\sqrt{2}$	-7	$(3 - \sqrt{2})$	13	R (4)	1/13
`inert3`	$3 + 3\sqrt{2}$	-9	(3)	5	R (4)	2/5
`d2_3`	$2 + 3\sqrt{2}$	-14	$(\sqrt{2}) \cdot (3 + \sqrt{2})$	13	R (5)	2/13
`dm2_3`	$-2 + 3\sqrt{2}$	-14	$(\sqrt{2}) \cdot (3 - \sqrt{2})$	15	R (5)	3/15
`p17a`	$-1 + 3\sqrt{2}$	-17	$(5 + 2\sqrt{2})$	8	R (6)	0/8
`p17b`	$1 + 3\sqrt{2}$	-17	$(5 - 2\sqrt{2})$	8	R (6)	1/8
`d0_3`	$3\sqrt{2}$	-18	$(\sqrt{2}) \cdot (3)$	6	R (6)	3/6

label	d	N(d)	(d)	entries	$n = 2$	isometric
p23a	$-3+4\sqrt{2}$	-23	$(5+\sqrt{2})$	6	R (5)	3/6
p23b	$3+4\sqrt{2}$	-23	$(5-\sqrt{2})$	8	R (6)	1/8
inert5	$5+5\sqrt{2}$	-25	(5)	5	R (6)	5/5
p31a	$-1+4\sqrt{2}$	-31	$(7+3\sqrt{2})$	5	N (9 roots)	–
p31b	$1+4\sqrt{2}$	-31	$(7-3\sqrt{2})$	5	R (8)	2/5
d4_5	$4+5\sqrt{2}$	-34	$(\sqrt{2})\cdot(5+2\sqrt{2})$	12	N (34 roots)	–
dm4_5	$-4+5\sqrt{2}$	-34	$(\sqrt{2})\cdot(5-2\sqrt{2})$	10	N (22 roots)	–
p41a	$-3+5\sqrt{2}$	-41	$(7+2\sqrt{2})$	2	R (12)	2/2
p41b	$3+5\sqrt{2}$	-41	$(7-2\sqrt{2})$	2	R (12)	2/2
d2_5	$2+5\sqrt{2}$	-46	$(\sqrt{2})\cdot(5+\sqrt{2})$	8	N (43 roots)	–
dm2_5	$-2+5\sqrt{2}$	-46	$(\sqrt{2})\cdot(5-\sqrt{2})$	6	N (45 roots)	–
p47a	$-5+6\sqrt{2}$	-47	$(7+\sqrt{2})$	6	R (10)	4/6
d7_7	$7+7\sqrt{2}$	-49	$(3+\sqrt{2})\cdot(3-\sqrt{2})$	23	R (5)	2/23
d0_5	$5\sqrt{2}$	-50	$(\sqrt{2})\cdot(5)$	2	N (44 roots)	–
d6_7	$6+7\sqrt{2}$	-62	$(\sqrt{2})\cdot(7+3\sqrt{2})$	8	N (101 roots)	–
dm6_7	$-6+7\sqrt{2}$	-62	$(\sqrt{2})\cdot(7-3\sqrt{2})$	4	R (11)	3/4
d3_6	$3+6\sqrt{2}$	-63	$(3)\cdot(3-\sqrt{2})$	24	R (8)	2/24
dm3_6	$-3+6\sqrt{2}$	-63	$(3)\cdot(3+\sqrt{2})$	23	N (21 roots)	–
d4_7	$4+7\sqrt{2}$	-82	$(\sqrt{2})\cdot(7+2\sqrt{2})$	3	N (94 roots)	–
dm4_7	$-4+7\sqrt{2}$	-82	$(\sqrt{2})\cdot(7-2\sqrt{2})$	3	N (48 roots)	–
d2_7	$2+7\sqrt{2}$	-94	$(\sqrt{2})\cdot(7+\sqrt{2})$	1	N (41 roots)	–
dm2_7	$-2+7\sqrt{2}$	-94	$(\sqrt{2})\cdot(7-\sqrt{2})$	4	N (57 roots)	–
d0_7	$7\sqrt{2}$	-98	$(\sqrt{2})\cdot(3+\sqrt{2})\cdot(3-\sqrt{2})$	23	R (8)	2/23
d3_8	$3+8\sqrt{2}$	-119	$(3+\sqrt{2})\cdot(5+2\sqrt{2})$	6	N (70 roots)	–
d9_10	$9+10\sqrt{2}$	-119	$(3+\sqrt{2})\cdot(5-2\sqrt{2})$	10	R (8)	3/10
dm3_8	$-3+8\sqrt{2}$	-119	$(3-\sqrt{2})\cdot(5-2\sqrt{2})$	8	N (15 roots)	–
dm9_10	$-9+10\sqrt{2}$	-119	$(3-\sqrt{2})\cdot(5+2\sqrt{2})$	6	R (12)	3/6
d6_9	$6+9\sqrt{2}$	-126	$(\sqrt{2})\cdot(3)\cdot(3+\sqrt{2})$	30	N (35 roots)	–
dm6_9	$-6+9\sqrt{2}$	-126	$(\sqrt{2})\cdot(3)\cdot(3-\sqrt{2})$	27	R (11)	5/27
d4_9	$4+9\sqrt{2}$	-146	$(\sqrt{2})\cdot(9+2\sqrt{2})$	2	N (154 roots)	–
dm4_9	$-4+9\sqrt{2}$	-146	$(\sqrt{2})\cdot(9-2\sqrt{2})$	3	N (106 roots)	–
dm2_9	$-2+9\sqrt{2}$	-158	$(\sqrt{2})\cdot(9-\sqrt{2})$	5	N (117 roots)	–
d1_9	$1+9\sqrt{2}$	-161	$(3-\sqrt{2})\cdot(5-\sqrt{2})$	4	R (18)	4/4
d9_11	$9+11\sqrt{2}$	-161	$(3-\sqrt{2})\cdot(5+\sqrt{2})$	3	N (52 roots)	–
dm1_9	$-1+9\sqrt{2}$	-161	$(3+\sqrt{2})\cdot(5+\sqrt{2})$	6	R (18)	3/6
dm9_11	$-9+11\sqrt{2}$	-161	$(3+\sqrt{2})\cdot(5-\sqrt{2})$	7	N (50 roots)	–
d4_11	$4+11\sqrt{2}$	-226	$(\sqrt{2})\cdot(11+2\sqrt{2})$	2	N (84 roots)	–
dm4_11	$-4+11\sqrt{2}$	-226	$(\sqrt{2})\cdot(11-2\sqrt{2})$	2	N (136 roots)	–
d10_13	$10+13\sqrt{2}$	-238	$(\sqrt{2})\cdot(3-\sqrt{2})\cdot(5-2\sqrt{2})$	9	N (83 roots)	–
d2_11	$2+11\sqrt{2}$	-238	$(\sqrt{2})\cdot(3-\sqrt{2})\cdot(5+2\sqrt{2})$	10	N (104 roots)	–
dm10_13	$-10+13\sqrt{2}$	-238	$(\sqrt{2})\cdot(3+\sqrt{2})\cdot(5+2\sqrt{2})$	10	N (165 roots)	–
dm2_11	$-2+11\sqrt{2}$	-238	$(\sqrt{2})\cdot(3+\sqrt{2})\cdot(5-2\sqrt{2})$	7	N (129 roots)	–
d1_12	$1+12\sqrt{2}$	-287	$(3+\sqrt{2})\cdot(7+2\sqrt{2})$	5	R (24)	5/5
dm1_12†	$-1+12\sqrt{2}$	-287	$(3-\sqrt{2})\cdot(7-2\sqrt{2})$	0	N (125 roots)	–
d12_15	$12+15\sqrt{2}$	-306	$(\sqrt{2})\cdot(3)\cdot(5+2\sqrt{2})$	3	N (122 roots)	–
dm12_15	$-12+15\sqrt{2}$	-306	$(\sqrt{2})\cdot(3)\cdot(5-2\sqrt{2})$	5	N (62 roots)	–
d16_17	$16+17\sqrt{2}$	-322	$(\sqrt{2})\cdot(3+\sqrt{2})\cdot(5+\sqrt{2})$	13	N (108 roots)	–
d4_13	$4+13\sqrt{2}$	-322	$(\sqrt{2})\cdot(3+\sqrt{2})\cdot(5-\sqrt{2})$	2	N (101 roots)	–
dm16_17	$-16+17\sqrt{2}$	-322	$(\sqrt{2})\cdot(3-\sqrt{2})\cdot(5-\sqrt{2})$	13	N (94 roots)	–
dm4_13	$-4+13\sqrt{2}$	-322	$(\sqrt{2})\cdot(3-\sqrt{2})\cdot(5+\sqrt{2})$	2	N (103 roots)	–
dm10_15	$-10+15\sqrt{2}$	-350	$(\sqrt{2})\cdot(5)\cdot(3-\sqrt{2})$	8	N (177 roots)	–
d6_15	$6+15\sqrt{2}$	-414	$(\sqrt{2})\cdot(3)\cdot(5+\sqrt{2})$	5	N (161 roots)	–
d12_17	$12+17\sqrt{2}$	-434	$(\sqrt{2})\cdot(3-\sqrt{2})\cdot(7-3\sqrt{2})$	2	N (158 roots)	–
d4_15	$4+15\sqrt{2}$	-434	$(\sqrt{2})\cdot(3-\sqrt{2})\cdot(7+3\sqrt{2})$	3	N (99 roots)	–
dm12_17	$-12+17\sqrt{2}$	-434	$(\sqrt{2})\cdot(3+\sqrt{2})\cdot(7+3\sqrt{2})$	3	N (154 roots)	–

label	d	N(d)	(d)	entries	$n = 2$	isometric
dm4_15	$-4 + 15\sqrt{2}$	-434	$(\sqrt{2}) \cdot (3 + \sqrt{2}) \cdot (7 - 3\sqrt{2})$	4	N (180 roots)	–
d21_21	$21 + 21\sqrt{2}$	-441	$(3) \cdot (3 + \sqrt{2}) \cdot (3 - \sqrt{2})$	4	N (133 roots)	–
d6_21	$6 + 21\sqrt{2}$	-846	$(\sqrt{2}) \cdot (3) \cdot (7 + \sqrt{2})$	3	N (125 roots)	–
d0_21	$21\sqrt{2}$	-882	$(\sqrt{2}) \cdot (3) \cdot (3 + \sqrt{2}) \cdot (3 - \sqrt{2})$	10	N (158 roots)	–
d28_35	$28 + 35\sqrt{2}$	-1666	$(\sqrt{2}) \cdot (3 + \sqrt{2}) \cdot (3 - \sqrt{2}) \cdot (5 + 2\sqrt{2})$	3	N (180 roots)	–
dm28_35	$-28 + 35\sqrt{2}$	-1666	$(\sqrt{2}) \cdot (3 + \sqrt{2}) \cdot (3 - \sqrt{2}) \cdot (5 - 2\sqrt{2})$	3	N (234 roots)	–

Table 4: The certificate at $n_0 = \mathcal{N}(d) + 1$ for each d: number of Vinberg roots used, number of vertex symmetries in the product g, and a rational interval containing a real eigenvalue > 1 of g.

label	n_0	roots	factors	eigenvalue in
unit	7	80	2	$[39/2, 20]$
ram2	6	40	1	$[21/2, 11]$
p7a	6	80	1	$[5/2, 3]$
p7b	5	33	1	$[19/2, 10]$
inert3	4	40	1	$[6, 13/2]$
d2_3	3	20	2	$[12, 25/2]$
dm2_3	4	40	2	$[13/2, 7]$
p17a	3	20	2	$[19/2, 10]$
p17b	3	20	2	$[75/2, 38]$
d0_3	3	33	2	$[113/2, 57]$
p23a	4	20	2	$[27/2, 14]$
p23b	3	20	1	$[27, 55/2]$
inert5	3	31	2	$[37/2, 19]$
p31a	2	9	2	$[607/2, 304]$
p31b	3	18	2	$[212, 425/2]$
d4_5	2	20	2	$[19/2, 10]$
dm4_5	2	20	1	$[205/2, 103]$
p41a	3	40	2	$[177/2, 89]$
p41b	3	40	2	$[131/2, 66]$
d2_5	2	20	1	$[19/2, 10]$
dm2_5	2	20	1	$[19/2, 10]$
p47a	3	20	2	$[10, 21/2]$
d7_7	3	80	1	$[11/2, 6]$
d0_5	2	40	1	$[21, 43/2]$
d6_7	2	20	2	$[89/2, 45]$
dm6_7	3	157	1	$[81/2, 41]$
d3_6	3	37	2	$[1643/2, 822]$
dm3_6	2	20	1	$[15, 31/2]$
d4_7	2	40	1	$[21, 43/2]$
dm4_7	2	48	1	$[131/2, 66]$
d2_7	2	40	2	$[205/2, 103]$
dm2_7	2	20	2	$[205/2, 103]$
d0_7	3	142	1	$[117/2, 59]$
d3_8	2	20	1	$[11/2, 6]$
d9_10	3	40	2	$[21, 43/2]$
dm3_8	2	15	2	$[3054, 6109/2]$
dm9_10	3	40	2	$[225/2, 113]$
d6_9	2	20	2	$[205/2, 103]$
dm6_9	3	160	2	$[39/2, 20]$
d4_9	2	40	1	$[19/2, 10]$
dm4_9	2	20	2	$[267/2, 134]$
dm2_9	2	20	2	$[5293, 10587/2]$

label	n_0	roots	factors	eigenvalue in
d1_9	3	157	2	$[16, 33/2]$
d9_11	2	40	1	$[39/2, 20]$
dm1_9	3	191	1	$[23/2, 12]$
dm9_11	2	40	1	$[25/2, 13]$
d4_11	2	80	2	$[11603/2, 5802]$
dm4_11	2	40	2	$[267/2, 134]$
d10_13	2	40	2	$[89/2, 45]$
d2_11	2	40	1	$[61, 123/2]$
dm10_13	2	40	2	$[1813/2, 907]$
dm2_11	2	40	1	$[23, 47/2]$
d1_12	3	160	1	$[49/2, 25]$
dm1_12	2	40	1	$[7, 15/2]$
d12_15	2	40	1	$[135/2, 68]$
dm12_15	2	62	2	$[205/2, 103]$
d16_17	2	80	1	$[21, 43/2]$
d4_13	2	80	1	$[126, 253/2]$
dm16_17	2	80	1	$[337/2, 169]$
dm4_13	2	80	1	$[679/2, 340]$
dm10_15	2	40	1	$[61, 123/2]$
d6_15	2	20	2	$[712, 1425/2]$
d12_17	2	40	2	$[2502, 5005/2]$
d4_15	2	80	2	$[51589, 103179/2]$
dm12_17	2	40	2	$[3395, 6791/2]$
dm4_15	2	20	1	$[23, 47/2]$
d21_21	2	40	1	$[69/2, 35]$
d6_21	2	80	1	$[712, 1425/2]$
d0_21	2	80	1	$[93/2, 47]$
d28_35	2	160	1	$[595/2, 298]$
dm28_35	2	80	1	$[262, 525/2]$

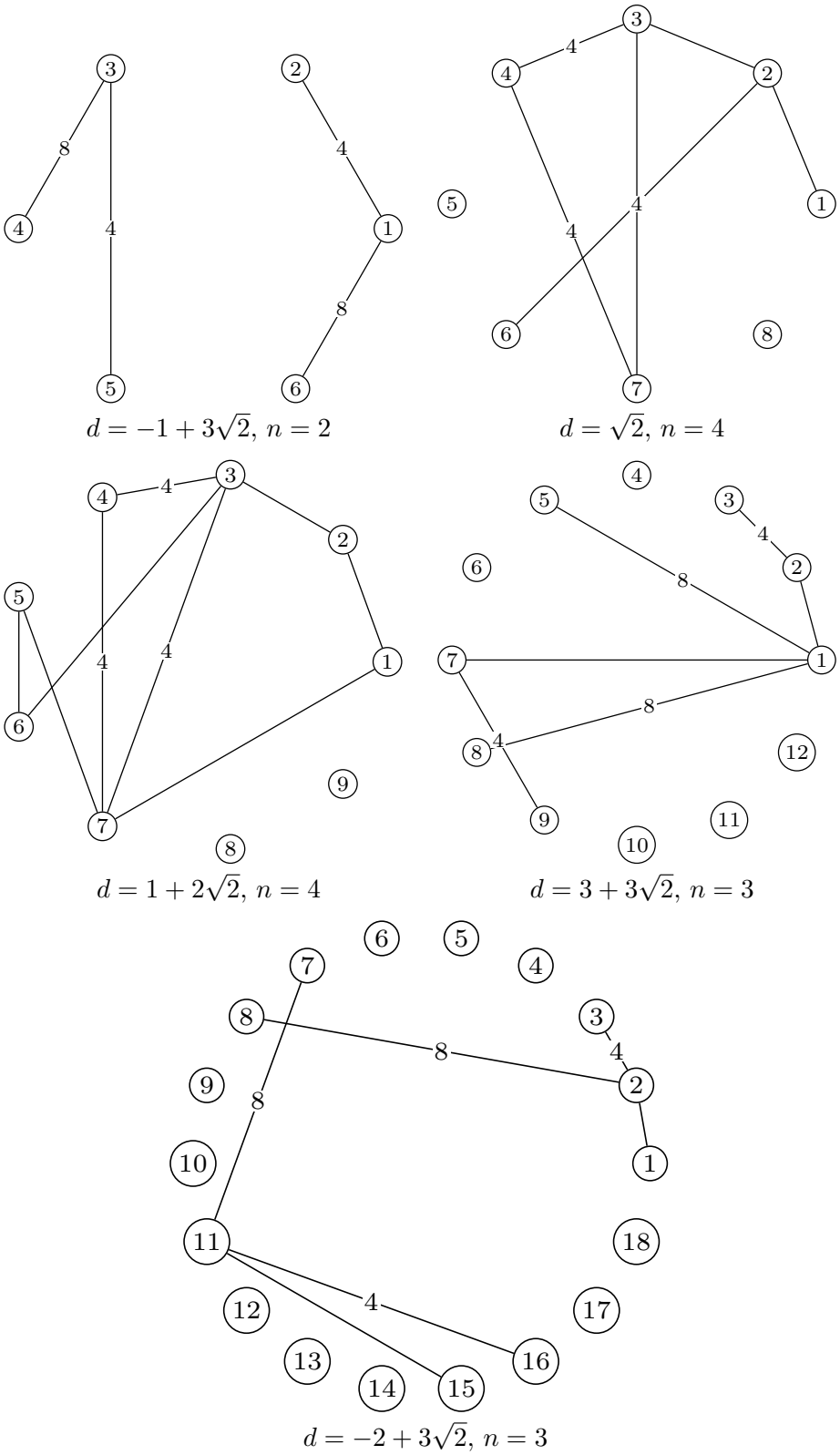


FIGURE 1. Coxeter diagrams of new reflective cases. Vertices are facets. No edge means angle $\pi/2$, an unlabelled edge $\pi/3$, and an edge labelled k means π/k. Ultra-parallel pairs (dotted edges) are omitted for legibility. The full Gram matrices are in the ancillary files. The diagram for $(\sqrt{2}, 5)$, with 27 facets, is given only there.